

Chapter 23

Polar and *wh*- question formation in Ekhwa Adara

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This paper presents the first description of polar and *wh*- interrogatives in Ekhwa Adara. In Ekhwa Adara, polar questions are optionally marked by sentence-initial Q particles and involve final-vowel lengthening and L% boundary tones, while *wh*-question formation involves optional *wh*- movement where the *wh*- item obligatorily precedes a predetermined focus particle. Long-distance *wh*- movement is possible and requires pronominal resumption. Partial *wh*- movement is also possible but seemingly does not require pronominal resumption. Indirect questions are formed via relativization, except when embedded under ‘ask’. Multiple *wh*- questions exhibit an absence of superiority effects and multiple *wh*- fronting might be available, albeit possibly constrained by an anti-superiority condition.

1 Introduction

Adara (ISO 639-3 [KAD]), also known as Eda, Edra, and Kadara, is an under-documented Benue-Congo language spoken by approximately 300,000 (Hon et al. 2018) to 500,000 people (Simons & Fennig 2018) in Kaduna and Niger states in Nigeria, as highlighted in Figure 1. Figure 2 provides a more detailed map of the Adara-speaking area of Nigeria taken from Hon et al. (2018), specifying the locations of the various Adara dialects.



Figure 1: Map of Nigeria showing the Adara-speaking region

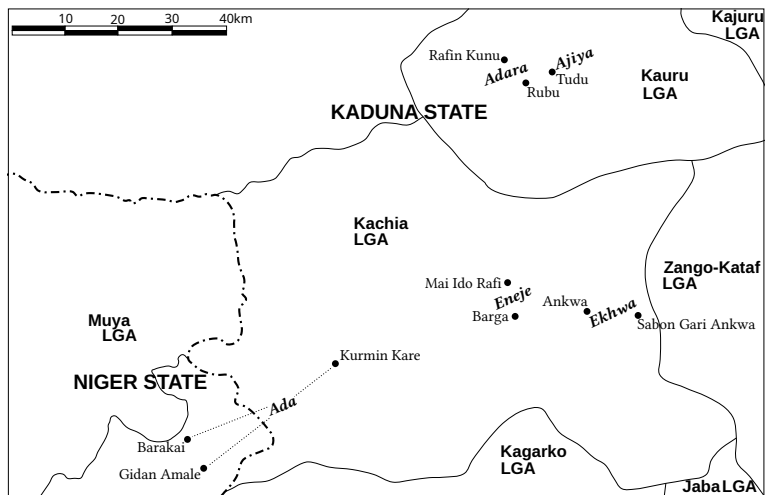


Figure 2: Map of the Adara-speaking area and its dialects (based on Hon et al. (2018))

Despite its large number of speakers, very little research has been done on Adara. This paper presents the first description of question formation in the language. Specifically, we investigate polar and *wh*- questions in Ekhwa Adara¹, the least researched of Adara's five dialects. Overall, we have found that polar questions in Ekhwa Adara involve final-vowel lengthening + L% boundary tones and that *wh*-movement in the language is optional. Where *wh*- movement does occur, the ex-situ *wh*- item obligatorily precedes a corresponding focus marker whose choice is determined by the animacy (human vs. non-human) and grammatical function (argument vs. adjunct) of the *wh*- item. Furthermore, both long-distance and partial *wh*- movement is possible in Ekhwa Adara. Lastly, and perhaps most remarkably, due to its (to the best of our knowledge) unattested status in the grammars of African languages, Ekhwa Adara may allow for multiple *wh*- fronting, but is possibly constrained by an anti-superiority condition.

This article is structured in the following way. In Section 2, we describe the formation of polar questions in Ekhwa Adara. Section 3 details our findings on *wh*- questions in Ekhwa Adara, including *wh*-in-situ phenomena, *wh*- movement (including partial and long-distance), embedded question formation, and multiple *wh*- questions. We conclude in Section 4 with a summary of our key findings and questions for future research.

2 Polar questions

Ekhwa Adara's primary polar question formation strategy is intonational in nature. Matrix polar questions involve final-vowel lengthening and L% tones, as is characteristic of the languages of the Sudanic belt region (Rialland 2007a, 2009, Cahill 2012, 2015). We see that the declarative sentences in (1a), (2a), and (3a) below all contain a short final vowel which is lengthened in their polar interrogative counterparts ((1b), (2b), and (3b), respectively).²

- (1) a. ijâ sù kɪ́ɔ́
 Ija PROG cry
 'Ija is crying.'

¹The data and judgments presented in this paper come exclusively from fieldwork with the third author, a native speaker of Ekhwa Adara, in the context of a Field Methods class at the CUNY Graduate Center in Fall 2022. Data are presented in IPA. The following diacritics are used to mark surface tone: $\acute{V}$ = high, $\hat{V}$ = low, $\bar{V}$ = mid; $\breve{V}$ = falling, $\tilde{V}$ = rising.

²In response to one reviewer, we are currently unaware of any other prosodic differences between declaratives and interrogatives in Ekhwa Adara.

- b. ijâ sù kɪɔ:
Ija PROG cry.Q
'Is Ija crying?'
- (2) a. omúsé ku ɔ́ó utébur
Omuse PFV buy table
'Omuse bought a table.'
- b. omúsé ku ɔ́ó utébû:r
Omuse PFV buy table.Q
'Did Omuse buy a table?'
- (3) a. omúsé ku kɪɔ ɔɪàì
Omuse PFV cry yesterday
'Omuse cried yesterday.'
- b. omúsé ku kɪɔ ɔɪàì:
Omuse PFV cry yesterday.Q
'Did Omuse cry yesterday?'

Furthermore, we see that intonational phrase-final high and mid tones, such as in (1a) *kɪɔ* 'cry' and (2a) *utébur* 'table', are both realized with falling pitch movements in their polar interrogative counterparts. In (1b), the phrase-final syllable *kɪɔ:* exhibits a falling tone and similarly in (2b), the final syllable of *utébû:r* is also realized with a falling tone. On the other hand, intonational phrase-final L tones such as in (3a) *ɔɪàì* 'yesterday' are still realized as L in their polar interrogative counterparts, as seen in (3b). Crucially, they are not realized with falling pitch movements. We also see that the sentences in (1) make use of the progressive aspect, whereas the sentences in (2) make use of the perfective aspect. The data therefore indicate that final-vowel lengthening and L% tone insertion are not tense/aspect dependent. Additionally, given that the final vowel quality in each example differs, we conclude that polar questions are formed by true vowel lengthening and not simply the addition of a uniform vowel, which occurs in some languages of the region (Rialland 2007b, Cahill 2015).

Syntactically, polar questions may be marked by a clause-initial Q particle *kó*, which seems to be borrowed from Hausa (Bawa 2023). While this Q particle is optional in matrix polar questions (4a) and *wh*-in-situ constructions (4b), it is obligatory in embedded polar questions (4c).

- (4) a. (kó) ijâ sù kɪɔ:
Q Ija PROG cry.Q
'Is Ija crying?'

- b. (kó) ijâ ku ɔ́ mčí
Q Ija PFV buy what
'What did Ija buy?'
- c. omúsé ku ɔ́:ru *(kó) ijâ ku kɔ́
Omuse PFV ask Q Ija PFV cry
'Omuse asked if Ija cried.'

From (4c), we also see that in embedded polar questions, where the Q particle obligatorily appears in the embedded clause to mark the scope of the indirect question, there is no final-vowel lengthening or L% boundary tone. Thus, final-vowel lengthening and insertion of L% boundary tones represent main clause phenomena in the language, as in Ikpana (Kandybowicz et al. 2023).

3 *Wh*- questions

3.1 *Wh*- in-situ

All argument *wh*- items may appear in-situ in root clauses.³ In (5), we observe the *wh*- item appearing in subject position (5a), direct object position (5b, 5d, 5e), and indirect object position (5c).⁴

- (5) a. iwé sú ɔ́ egbé
who FUT buy house
'Who will buy a house?'
- b. ijâ ku ɔ́ mčí/iwé
Ija PFV see what/who
'What/who did Ija see?'
- c. ijâ ku dʒe iwé onsě
Ija PFV give who name
'Who did Ija give a name to?'
- d. ijâ ku dʒe awé-̀̀̀ mčí
Ija PFV give child-DEF what
'What did Ija give the child?'

³Though as discussed below, it is not clear whether focused subjects are in-situ or ex-situ at this stage of our research.

⁴From (5c, 5d), we observe that in Ekhwa Adara double object constructions, the indirect object immediately precedes the direct object.

- e. ijâ ku ɔ́ ocmá egbé
 Ija PFV buy which house
 ‘Which house did Ija buy?’

As previously discussed in Section 2, the Q particle *kó* may optionally appear in root clause *wh*- in-situ questions. Currently, we only have evidence of its (optional) appearance in object root clause *wh*- in-situ questions such as (6a, 6b). We lack the data to determine whether Q may also appear in non-object root *wh*- in-situ constructions, but we suspect this to be the case.⁵ Note that in (6b), we observe a focused *wh*- subject after *kó*. This, however, should not be considered an instance where *kó* is marking an ex-situ *wh*- subject question, because Q-marking does not occur in ex-situ *wh*- questions (note the absence of the Q particle in the data presented in Section 3.2). Rather, in this sentence, *kó* is marking the in-situ *wh*- object *mcí* ‘what’. Given our consultant’s emphasis on beginning an interrogative sentence with some indication of intent (either through the Q particle *kó* or an initial focused *wh*-phrase), the optional *kó* marker most likely signals that there is (at least) one in-situ interrogative expression following the subject.

- (6) a. (kó) ijâ ku ɔ́ mcí
 Q Ija PFV buy what
 ‘What did Ija buy?’
 b. (kó) iwé ŋu ku ɔ́ mcí
 Q who FOC PFV buy what
 ‘Who bought what?’

All adjunct *wh*- items may also appear in-situ in root clauses. This is demonstrated in (7) for a variety of adjuncts (locative (7a), temporal (7b), manner/quantity (7c) and reason (7d)).

- (7) a. ijâ ku ɔ́ egbé imbí
 Ija PFV buy house where
 ‘Where did Ija buy a house?’
 b. ijâ ku ɔ́ egbé ocmá atú/apa
 Ija PFV buy house which day/time
 ‘When (i.e., which day/time) did Ija buy a house?’

⁵Our consultant seems to prefer either ex-situ or focus marked in-situ *wh*- subject questions, at least in root contexts. This may explain why the Q particle is limited to *wh*- object questions in our dataset.

- c. a sù nɛ otúmá-ŋ níní
 3SG.NOM PROG do work-DEF how
 ‘How is s/he doing the work?’
 ‘For how much (money) is s/he doing the work?’
- d. ijâ ku ɔ́ ɛgbé-ŋ domín incí
 Ija PFV buy house-DEF reason what
 ‘Why did Ija buy the house?’

Long-distance *wh*- in-situ is also attested. All *wh*- items may appear in-situ in embedded clausal complements and take wide scope over the matrix clause. We observe this in (8), where the *wh*- items *iwé* ‘who’ (8a) and *incí* ‘what’ (8b) appear as the subject and object of the embedded clause, respectively. Crucially, these sentences cannot receive indirect question interpretations.⁶ Instead, the sentences are interpreted as long-distance in-situ interrogatives, with the *wh*- item in the complement clause taking matrix scope.⁷

- (8) a. omúsé ga iwé ku ɔ́ ɛgbé
 Omuse say who PFV buy house
 ‘Who did Omuse say bought a house?’
 NOT: ‘Omuse said who bought a house.’
- b. omúsé ga ijâ ku ɔ́ incí
 Omuse say Ija PFV buy what
 ‘What did Omuse say Ija bought?’
 NOT: ‘Omuse said what Ija bought.’

Embedded in-situ adjunct *wh*- expressions may only be interpreted as originating in the embedded clause. We demonstrate this in (9), where the *wh*- items ‘where,’ ‘when,’ ‘how much,’ and ‘why’ can only be interpreted as being generated in the embedded clause, that is, modifying the verb ɔ́ ‘buy’, as opposed to the matrix verb *ga* ‘say’.

- (9) a. omúsé ga ijâ ku ɔ́ ɛgbé imbí
 Omuse say Ija PFV buy house where
 ‘Where did Omuse say Ija bought a house?’

⁶As discussed in (23) below, this would require relativization.

⁷The Q particle *kó* is unavailable in this construction to mark the scope of the *wh*- element. In fact, with the Q particle *kó* in sentence initial position, the *wh*- questions in (8) become quotative polar questions, expressing ‘Did Omuse say: “Who bought a house?”’ and ‘Did Omuse say: “What did Ija buy?”’ respectively.

- ✓ ‘Where-buy’
 * ‘Where-say’
- b. omúsé ga ijâ ku ɔ́ egbé ocmá apa
 Omuse say Ija PFV buy house which time
 ‘When did Omuse say Ija bought a house?’
 ✓ ‘When-buy’
 * ‘When-say’
- c. omúsé ga ijâ ku ɔ́ egbé níní
 Omuse say Ija PFV buy house how
 ‘For how much did Omuse say Ija bought a house?’
 ✓ ‘For how much-buy’
 * ‘For how much-say’
- d. omúsé ga ijâ ku ɔ́ egbé domín mčí
 Omuse say Ija PFV buy house reason what
 ‘Why did Omuse say Ija bought a house?’
 ✓ ‘Why-buy’
 * ‘Why-say’

3.2 *Wh*- movement

3.2.1 Focus particles

Ekhwa Adara is an optional *wh*- movement language. As we saw in Section 3.1, all *wh*- arguments and adjuncts may remain in-situ with or without the initial Q particle *kó*. The interrogative phrase, however, may also be fronted to clause-initial position. This optional *wh*-movement requires a constituent focus particle, which immediately follows the *wh*- item and varies according to the animacy and type of the interrogative expression. When undergoing movement, we see that the human interrogative expression *iwé* ‘who’ must be followed by the particle *ɲu* (10a)⁸, the non-human interrogative expression *mčí* ‘what’ must be followed by the particle *mo* (10b), and finally, the D-linked interrogative *ocmá* (10c) ‘which’ and interrogative adjuncts such as *imbí* ‘where’ (10d), *ocmá apa* ‘when’ (10e), and *níní* ‘how/for how much’ (10f) must be followed by the particle *ku*. The only

⁸Unlike with long-distance *wh*- movement, there is no direct evidence in cases of monoclausal *wh*- questions that the subject is moved at all, since *wh*- items may be accompanied by a focus marker even when left in-situ, as shown in (16). There also seems to be a preference for the focus-marked construction when the interrogative phrase is the subject, while the same is not true for other arguments. Our point, in presenting (10a), is merely that, if moved, the *wh*-subject should behave like non-subject *wh*- items and require the focus particle.

exception is the *wh*- adjunct *domín mčí* ‘why’ (literally, ‘what reason’) which may either precede *ku*, like all other *wh*- adjuncts, or *mo*, like *mčí* ‘what’ (10g).

- (10) a. iwé *(*ɲu*) — ku ɔ́ eǵbé
 who FOC PFV buy house
 ‘Who bought a house?’
 b. mčí *(*mo*) ijâ ku ɔ́ —
 what FOC Ija PFV buy
 ‘What did Ija buy?’
 c. ocná eǵbé *(*ku*) ijâ ku ɔ́ —
 which house FOC Ija PFV buy
 ‘Which house did Ija buy?’
 d. mbí *(*ku*) ijâ ku ɔ́ eǵbé —
 what FOC Ija PFV buy house
 ‘Where did Ija buy a house?’
 e. ocná apa *(*ku*) ijâ ku ɔ́ eǵbé —
 which time FOC Ija PFV buy house
 ‘When did Ija buy a house?’
 f. níní *(*ku*) a sù nɛ otúmá-ɲ —
 how FOC 3SG.NOM PROG do work-DEF
 ‘How/for how much is s/he doing the work?’
 g. domín mčí *(*ku/mo*) ijâ ku ɔ́ eǵbé —
 reason what FOC Ija PFV buy house
 ‘Why did Ija buy a house?’

The ungrammaticality of the examples in (11), in which the focus particles that follow the *wh*- phrase are switched, demonstrate that these markers are compatible only with specific *wh*- items – human *iwé* ‘who’ must be paired with *ɲu*, rather than *mo* or *ku*; non-human *mčí* ‘what’ must be paired with *mo*, rather than *ɲu* or *ku*; lastly, the D-linked interrogative *ocná* ‘which’ and *wh*- adjuncts, with the exception of *domín mčí* ‘why’, must be paired with *ku*, rather than *ɲu* or *mo*.

- (11) a. * iwé mo/ku — ku ɔ́ eǵbé
 who FOC PFV buy house
 Intended: ‘Who bought a house?’

- b. * *mcí* *ɲu*/ku *ijâ* ku *ɔ́* —
 what FOC Ija PFV buy
 Intended: ‘What did Ija buy?’
- c. * *ocná* *egbé* *ɲu*/mo *ijâ* ku *ɔ́* —
 which house FOC Ija PFV buy
 Intended: ‘Which house did Ija buy?’
- d. * *mbí* *ɲu*/mo *ijâ* ku *ɔ́* *egbé* —
 what FOC Ija PFV buy house
 Intended: ‘Where did Ija buy a house?’
- e. * *ocná* *apa* *ɲu*/mo *ijâ* ku *ɔ́* *egbé* —
 which time FOC Ija PFV buy house
 Intended: ‘When did Ija buy a house?’
- f. * *níní* *ɲu*/mo a *sù* *nɛ* *otúmá-ɲ* —
 how FOC 3SG.NOM PROG do work-DEF
 Intended: ‘How/for how much is s/he doing the work?’
- g. * *domín* *mcí* *ɲu* *ijâ* ku *ɔ́* *egbé* —
 reason what FOC Ija PFV buy house
 Intended: ‘Why did Ija buy a house?’

Furthermore, the examples in (12) show that the choice of *ɲu* or *mo* is truly dependent on the interrogative item itself and not on its syntactic relationship to the verb (e.g., case). As we can see, ex-situ *iwé* still requires *ɲu* when thematically linked to non-subject positions like direct objects (12a) and possessors of objects (12b). Likewise, ex-situ *mcí* selects the marker *mo* even when linked to the subject position (12c).

- (12) a. *iwé* *ɲu* *ijâ* ku *dá* —
 who FOC Ija PFV touch
 ‘Who has Ija touched?’
- b. *iwé* *ɲu* *ijâ* ku *ɔ́* *egbé-ɲ* —
 who FOC Ija PFV buy house-DEF
 ‘Who did Ija buy a house from?’
- c. *mcí* *mo* — ku *dá* *ijâ*
 what FOC PFV touch Ija
 ‘What has touched Ija?’

In addition to appearing in ex-situ *wh-* questions, these constituent focus particles may occur in answer fragments. Here, we see that (13b) is a grammatical response to (13a).

- (13) a. iwé ɲu mín
 who FOC here
 ‘Who is this?’
 b. omúsé ɲu
 Omuse FOC
 ‘(This is) Omuse.’

Both *ɲu* and *ku*⁹ also appear as focus particles in declarative sentences with focus fronting. Like in ex-situ interrogative constructions, focus markers follow left peripheral foci and the realization of the surfacing focus particle depends on the constituent it focus-marks. Consider the facts in (14) and (15), which illustrate.¹⁰

- (14) a. ijâ ɲu/*ku ɔ́ egbé
 Ija FOC buy house
 ‘IJA bought a house.’
 b. ijâ ɲu/*ku omúsé sú mani — ɔ́nfó
 Ija FOC Omuse FUT teach song
 ‘Omuse will teach a song (to) IJA.’

⁹The constituent focus particles *ɲu* and *ku* also occur in allomorphic variation with *ɲo* and *ko*, respectively. Further research is necessary to see whether this variation is phonologically conditioned. Note that the constituent focus particle variant *ko*, which bears a mid tone, is tonally (not segmentally) contrastive with the high tone-bearing initial interrogative particle *kó*.

¹⁰We initially glossed (14-15) with English *it*-cleft translations (e.g., (14a) was glossed as ‘It is IJA who bought a house’). Thanks to one of our reviewers, however, we realized that these translations might not be semantically appropriate given that *it*-cleft sentences in English have exhaustive properties which the sentences in (14-15) do not possess. For example, our consultant notes that in the case where Omuse will teach both Ija and Awen a song, (14b) can only be used if the hearer is unaware of Awen’s existence. We assume this does not constitute an exhaustive interpretation since Awen’s existence is not part of the common ground shared by both speakers. Thus, Awen is not in the domain of individuals whose “learning of the song from Omuse” we need to negate. An exhaustive interpretation of (14b) is only available when both Ija and Awen are relevant to the context (i.e., in the common ground) but only Ija will be taught the song (possibly among other students who, similar to Awen in the previous case, are not part of the common ground and thus, the alternative propositions involving them need not be negated).

- c. ená ŋu/??ku omúsé sú mani — ɔŋfó
 cow FOC Omuse FUT teach song
 ‘Omuse will teach a song to a cow.’
- (15) a. ɔŋfó ku/*ŋu omúsé sú mani-ŋ ijâ —
 song FOC Omuse FUT teach-3SG.ACC Ija
 ‘Omuse will teach Ija a SONG.’
- b. egbé ku/*ŋu/*mo ijâ ku ɔ́ —
 house FOC Ija PFV buy
 ‘Ija bought a HOUSE.’

The data in (14) and (15) suggest that the choice of focus marker in declarative sentences is related to animacy – focused animate DPs appear with *ŋu* (14), while focused inanimates are accompanied by *ku* (15). As before with ex-situ *wh*-questions, the choice of either *ŋu* or *ku* in declarative sentences does not seem to depend on the relationship of the moved phrase to other elements of the sentence (e.g., case, thematic role, etc.).

The focus markers introduced in this section are not limited to contexts in which an interrogative undergoes movement. *Wh*-items that remain in-situ may optionally precede these same particles (16).

- (16) a. ijâ ku dʒe iwé (ŋu) onsě
 Ija PFV give who FOC name
 ‘Who did Ija give a name to?’
- b. ijâ ku ɿ mɿ́ (mo)
 Ija PFV see what FOC
 ‘What did Ija see?’
- c. ijâ ku ɔ́ egbé mbí (ku)
 Ija PFV buy house where FOC
 ‘Where did Ija buy a house?’
- d. ijâ ku ɔ́ egbé ocíná apa (ku)
 Ija PFV buy house which time FOC
 ‘When did Ija buy a house?’
- e. a sù nɛ otúma-ŋ níní (ku)
 3SG.NOM PROG do work-DEF how FOC
 ‘How/for how much is s/he doing the work?’

- f. *ijâ ku ɔ́ egbé-ŋ domín incí (ku/mo)*
 Ija PFV buy house-DEF reason what FOC
 ‘Why did Ija buy the house?’

A list of Ekhwa Adara *wh*- items and the focus particles they are compatible with is presented in Table 1 below.

Table 1: Ekhwa Adara *wh*- items and their accompanying focus particles

<i>Wh</i> - item	Accompanying focus particle
<i>iwé</i> ‘who’	<i>ŋu</i>
<i>incí</i> ‘what’	<i>mo</i>
<i>imbí</i> ‘where’	<i>ku</i>
<i>ocíná</i> ‘which’	<i>ku</i>
<i>ocíná atú/apa</i> ‘when’	<i>ku</i>
<i>níní</i> ‘how’	<i>ku</i>
<i>domín incí</i> ‘why’	<i>ku/mo</i>

3.2.2 Long-distance movement

Long-distance *wh*- movement in Ekhwa Adara exhibits the same dependency between the moved phrase and its accompanying focus marker. Additionally, however, we see an asymmetry between subjects and non-subjects. The subject of an embedded clause, when fronted to sentence-initial position, must be resumed by an agreeing pronominal (17a), while resumption of displaced embedded objects is ungrammatical (17b).¹¹ This is typical of languages in the region, such as Nupe (Kandybowicz 2004, 2008) and Igbo (Amaechi & Georgi 2019).

- (17) a. *iwé ŋu omúsé ga *(a) ku ɔ́ egbé*
 who FOC Omuse say 3SG.NOM PFV buy house
 ‘Who did Omuse say bought a house?’
 b. *incí mo omúsé ga ijâ ku ɔ́ (*ŋ)*
 what FOC Omuse say Ija PFV buy 3SG.ACC
 ‘What did Omuse say Ija bought?’

¹¹We did not observe resumption in the indirect object or adjunct positions in long-distance *wh*-movement. Furthermore, to the best of our knowledge, the language does not have any null resumptive forms. That is, it only has overt pronominals.

Examples (18a)-(18c) demonstrate that movement of a *wh*- adjunct in biclausal questions does not require pronominal resumption or any other indication of the position in which the *wh*- adjunct was generated. Accordingly, the sentences may receive either matrix or embedded interpretations, questioning the saying event or the buying event, respectively. This is in contrast to examples containing in-situ *wh*- adjuncts, as well as in-situ *wh*- arguments. As we saw earlier in (9), long-distance (i.e., embedded) *wh*- in-situ with adjuncts forces embedded interpretations, meaning that the *wh*- phrase can only be interpreted as modifying the lower clause verb, whereas in (8), long-distance *wh*- in-situ with arguments forces matrix interpretations, meaning that such sentences must be interpreted as long-distance in-situ *wh*- interrogatives as opposed to indirect questions.

- (18) a. *mbí ku omúsé ga ijâ ku ɔ́ egbé*
 where FOC Omuse say Ija PFV buy house
 ‘Where did Omuse say Ija bought a house?’
 ✓‘Where-buy’
 ✓‘Where-say’
- b. *ocná apa ku omúsé ga ijâ ku ɔ́ egbé*
 which time FOC Omuse say Ija PFV buy house
 ‘When did Omuse say Ija bought a house?’
 ✓‘When-buy’
 ✓‘When-say’
- c. *domín mčí ku/mo omúsé ga ijâ ku ɔ́ egbé*
 reason what FOC Omuse say Ija PFV buy house
 ‘Why did Omuse say Ija bought a house?’
 ✓‘Why-buy’
 ✓‘Why-say’

3.2.3 Partial movement

Ekhwa Adara also allows partial *wh*- movement, that is, movement of an embedded interrogative to the initial position of a lower clause. When partially moved, *wh*- is accompanied by its corresponding focus particle (as discussed in Section 3.2.1). Unlike in long-distance *wh*- questions, partially moved subjects in Ekhwa Adara do not seem to require pronominal resumption (19a), erasing the subject–non-subject asymmetry that was exhibited in (17a, 17b) above.

- (19) a. *omúsé ga iwé ŋu — ku ɔ́ egbé*
 Omuse say who FOC PFV buy house
 ‘Who did Omuse say bought a house?’

- b. omúsé ga incí mo ijâ ku ɔ́ —
 Omuse say what FOC Ija PFV buy
 ‘What did Omuse say Ija bought?’
- c. omúsé ga imbí ku ijâ ku ɔ́ egbé —
 Omuse say where FOC Ija PFV buy house
 ‘Where did Omuse say Ija bought a house?’
 ✓ ‘Where-buy’
 * ‘Where-say’
- d. omúsé ga ocná apa ku ijâ ku ɔ́ egbé —
 Omuse say which time FOC Ija PFV buy house
 ‘When did Omuse say Ija bought a house?’
 ✓ ‘When-buy’
 * ‘When-say’
- e. omúsé ga níní ku ijâ ku ɔ́ egbé —
 Omuse say how FOC Ija PFV buy house
 ‘For how much did Omuse say Ija bought a house?’
 ✓ ‘How much-buy’
 * ‘How much-say’
- f. omúsé ga domín incí mo ijâ ku ɔ́ egbé —
 Omuse say reason what FOC Ija PFV buy house
 ‘Why did Omuse say Ija bought a house?’
 ✓ ‘Why-buy’
 * ‘Why-say’

As with embedded in-situ questions, partial movement of the *wh-* adjuncts in (19c)–(19f) can only be interpreted as originating in the immediately-containing clause (at least in biclausal contexts). Questions formed in this way do not require long-distance movement to sentence-initial position to mark scope, nor do they require (or allow) the clause-initial Q marker *kó*. We can thus classify Ekhwa Adara as a Simple/Naked partial *wh-* movement language, referencing Fanselow 2006’s typology. In this way, partial *wh-* movement in Ekhwa Adara resembles partial *wh-* fronting in other West African languages such as Krachi (Torrence & Kandybowicz 2013, 2015, Kandybowicz & Torrence 2015); Bono and Wasa (Kandybowicz 2017, 2020, Kandybowicz & Torrence 2015); Avatime (Devlin et al. 2021); and Ikpana (Kandybowicz et al. 2021, 2023).

Interestingly, there is seemingly a limit on how far a *wh-* item can move in partial movement structures. In (20), where there are two levels of embedding, we

see that *wh*- items can only either partially move within the originating clause (20a) without the resumptive pronoun or move long-distance to the matrix operator position (20b) with the resumptive pronoun. They cannot move to any intermediate position. That is, a *wh*- item originating in the lowest embedded clause cannot move to the left periphery of an intermediate clause, with or without a resumptive pronoun (20c).

- (20) a. awên ga (aɪ) ijâ ga-ŋ (aɪ) iwé ɲu — ku ɔ́ egbe
 Awen say COMP Ija say-3SG.ACC COMP who FOC PFV buy house
 ‘Who did Awen say that Ija said bought a house?’
 Lit: ‘Who did Awen say that Ija told her (Awen) bought a house?’
- b. iwé ɲu awên ga (aɪ) ijâ ga-ŋ (aɪ) a ku ɔ́
 who FOC Awen say COMP Ija say-3SG.ACC COMP 3SG.NOM PFV buy
 egbe
 house
 ‘Who did Awen say that Ija said bought a house?’
 Lit: ‘Who did Awen say that Ija told her (Awen) bought a house?’
- c. *awên ga (aɪ) iwé ɲu ijâ ga-ŋ (aɪ) (a) ku ɔ́
 Awen say COMP who FOC Ija say-3SG.ACC COMP 3SG.NOM PFV buy
 egbe
 house
 Intended: ‘Who did Awen say that Ija said bought a house?’

3.3 Embedded questions

Wh- movement to the left periphery of the lower clause also occurs in the formation of embedded questions under the matrix verb ‘ask’ (21).¹² This construction is identical to the partial *wh*- movement construction described above – both subjects and objects are moved without use of a resumptive pronoun and the clause-initial Q particle is absent. The scope of the *wh*- item in this case is limited to the lower clause due (presumably) to the semantic properties of ‘ask.’

¹²The lower clauses in (21) are indeed embedded questions under the matrix verb ‘ask’, rather than, as per a reviewer’s concern, quotations. In (i), the embedded first-person object refers to the speaker, as expected of an embedded question interpretation, rather than the matrix subject Omuse, which is what we would expect of a quotative interpretation.

(i) omúsé_i ku ɔ́:ru iwé ɲu — ri-i_{sp/*i}
 Omuse PFV ask who FOC see-1SG.ACC
 ‘Omuse_i asked who saw me_{sp/*i}’

- (21) a. omúsé ku ɔ́:ru iwé ɲu — ku ɔ́ egbé
Omuse PFV ask who FOC PFV buy house
‘Omuse asked who bought a house.’
b. omúsé ku ɔ́:ru incí mo ijâ ku ɔ́ —
Omuse PFV ask what FOC Ija PFV buy
‘Omuse asked what Ija bought.’

Indirect questions with ‘ask’ do not, however, require *wh*- movement in the embedded clause. (22) shows an example of an embedded *wh*- in-situ question under ‘ask’.

- (22) omúsé ku ɔ́:ru-i ijâ ku ɔ́ incí mo
Omuse PFV ask-1SG Ija PFV buy what FOC
‘Omuse asked me what Ija bought.’¹³

Indirect question complements of other embedding verbs are formed via relativization, generating what can be described as “concealed questions”. The sentence in (23a) is nearly identical to that in (21a), apart from the matrix verb and aspect marking¹⁴; however, only (21a) is grammatical. Local fronting of the *wh*-item within the lower clause does not achieve the intended reading in (23a) – i.e., it is not interpreted as an indirect question.¹⁵ In order to attain the intended reading, it is necessary to use a relative clause construction. In the case of an embedded ‘who’ question, the relative clause is headed by *anú-n* ‘the person’ (23b), replacing the *wh*- phrase *iwé ɲu* ‘who’. In the case of an indirect ‘what’ question, the relative clause is headed by *ɔŋǎ-ḡ* ‘the thing’ (23c) replacing *incí mo* ‘what’.¹⁶

¹³Our consultant notes that in (22), without the matrix first-person object *i*, the speaker would expect an answer (except in certain contexts), which defeats the purpose of the reported speech we are aiming to document.

¹⁴In many cases, we found a preference for certain sentences to be produced without overt aspect marking, even if the corresponding sentence with a marker was judged to be grammatical. Absence of the aspectual element in (23a), therefore, would not be expected to negatively affect its acceptability. If anything, we might expect it to improve it.

¹⁵Currently, we are unsure whether (23a) can be parsed as an instance of partial *wh*- movement, with the moved phrase taking scope over the matrix clause. Given that ‘know’ can embed a clause (see (i) in footnote 15 below), if the partial *wh*- parse is not available, this may suggest that ‘know’ predicates in Ekhwa Adara can form factive islands.

¹⁶Although the verb ‘know’ is followed by a noun phrase in every example in (23), we do not analyze *képéi* ‘know’ as an acquaintance verb similar to French *connaître* because it can be followed by a clause, as seen in (i).

- (i) omúsé k̂péi aɪ ijâ ɔ́ egbé
Omuse know COMP Ija buy house
‘Omuse knows that Ija bought a house.’

As indicated below, these structures translate more literally as ‘the person/thing that...’. Subject and object relative clauses of this type are both generally grammatical in Ekhwa Adara, using this exact construction.

- (23) a. *omúsé kǫ̀pé.ɾi iwé ɲu ku ɾó egbé
 Omuse know who FOC PFV buy house
 Intended: ‘Omuse knows who bought a house.’
- b. omúsé kǫ̀pé.ɾi anú-̀̀n da ku ɾó egbé
 Omuse know person-DEF REL PFV buy house
 ‘Omuse knows who bought a house.’
 Literally: ‘Omuse knows the person who bought a house.’
- c. omúsé kǫ̀pé.ɾi ɔŋǵǵ-̀̀ɲ da ijâ ku ɾó
 Omuse know thing-DEF REL Ija PFV buy
 ‘Omuse knows what Ija bought.’
 Literally: ‘Omuse knows the thing that Ija bought.’

This limitation of true embedded questions to ‘ask’ complement clauses connects Ekhwa Adara to Krachi (Torrence & Kandybowicz 2015), where identical facts obtain.

3.4 Multiple *wh*- questions

At present, we have observed three potential strategies for the formation of multiple *wh*- questions in Ekhwa Adara: (i) all *wh*- items remain in-situ; (ii) one *wh*- item is fronted to clause-initial position, while the other(s)¹⁷ remain(s) in-situ; and (iii) all *wh*- items are fronted (i.e., multiple *wh*- fronting). As expected, focus particles appearing in each of these three construction types must be the ones selected by the accompanying *wh*- item. We are certain that strategies (i) and (ii) exist in the grammar. We are less certain about the reality of strategy (iii) given unstable judgments observed over a one-year period, which we discuss below.

Beginning with strategy (i), example (24) demonstrates that multiple *wh*- questions may be formed simply by leaving all interrogative expressions in-situ.

- (24) (*kó) omúsé sú mani iwé (ɲu) mčí (mo)
 Q Omuse FUT teach who FOC what FOC
 ‘What will Omuse teach to who?’

¹⁷We do not have any data pertaining to questions involving more than two *wh*- elements, though we suspect that such questions would also use this strategy.

Note that multiple *wh*- questions formed in this way may not be marked by the clause-initial Q particle *kó*¹⁸, despite the fact that the Q marker may optionally appear when a single *wh*- item is left in-situ (6). Although focus markers may optionally accompany each in-situ interrogative expression, there is a preference for realizing the focus particle after the first *wh*- item, but not the second.

Turning next to strategy (ii), example (25) demonstrates that multiple *wh*- questions formed by moving a single interrogative item may optionally be marked by the Q particle *kó*, unlike certain other types of *wh*- movement constructions in Ekhwa Adara discussed in this paper (e.g., long-distance *wh*- movement, partial *wh*- movement, and possibly multiple *wh*- fronting, as described below). In example (25), we also observe that the hierarchically inferior *wh*- item (the object ‘what’) is fronted, crossing over the higher in-situ interrogative (the subject ‘who’), an apparent violation of the superiority condition (Kuno & Robinson 1972, Chomsky 1973, 1977).

- (25) (kó) incí mo iwé ku ɔ́ —
 Q what FOC who PFV buy
 ‘Who bought what?’

Similar behavior in the distribution of moved interrogative phrases in multiple *wh*- fronting constructions (see below) also suggests an absence of superiority effects in the language. In this way, Ekhwa Adara is similar to a number of other West African languages that have been shown to lack superiority effects outright (Saah 1994, Adesola 2005, 2006, Torrence & Kandybowicz 2015, Kandybowicz et al. 2023, Schurr et al. 2024).

The sentences in (26) below demonstrate strategy (iii) – fronting of multiple *wh*- phrases.¹⁹ When more than one *wh*- item is moved to initial position²⁰, the items are pronounced in an order inverse to the hierarchical position of their extraction sites.

¹⁸A sequence of two clause-initial occurrences of *kó* (i.e., one for each in-situ *wh*- item) is also prohibited in structures like (24).

¹⁹Since focused in-situ subjects are indistinguishable from their moved counterparts, we avoid presenting multiple *wh*- movement structures in which one interrogative is the subject. We attempted to elicit multiple *wh*- fronting structures in which one of the moved elements was an adjunct, but the types of marginal judgements that we received for these require further investigation.

²⁰We have only collected monoclausal multiple *wh*- questions so far. ‘Initial’ here, therefore, means both clause- and sentence-initial, but we have not yet determined that this must be the case.

- (26) a. omúsé sú mani ijâ ɔɲfó
 Omuse FUT teach Ija song
 ‘Omuse will teach Ija a song.’
 b. mčí_i mo iwé_j ɲu omúsé sú mani-*(ɲ)_j t_i
 what FOC who FOC Omuse FUT teach-3SG.ACC
 ‘What will Omuse teach to who?’
 c. * iwé_j ɲu mčí_i mo omúsé sú mani-*(ɲ)_j t_i
 who FOC what FOC Omuse FUT teach-3SG.ACC
 Intended: ‘What will Omuse teach to who?’

The canonical ordering of the two lower arguments in a double-object construction places the indirect object in a position immediately preceding the direct object (26a). When both of these arguments are fronted, as in (26b), however, the direct object interrogative (‘what’) must precede the indirect object (‘who’). The reverse order, which would both respect the superiority condition and remain closer in form to the canonical Ekhwa Adara word order, was consistently judged ungrammatical (26c). Assuming the accuracy of these judgment contrasts, we therefore observe a kind of anti-superiority effect in Ekhwa Adara multiple *wh*- movement constructions – the initial fronted *wh*- item must be hierarchically inferior to the other(s). The acceptability of sentences like (26b) might also suggest a recursive FocP domain in the Ekhwa Adara left periphery (contra Rizzi (1997)), unlike what we see in other African languages (e.g., Aboh 2004). However, the availability of focus-marking on in-situ *wh*- elements in the language (see (16)) likely supports an alternative analysis in which the peripheral focus particles that surface in multiple *wh*- fronting structures are more accurately “term focus” markers.

Further research is needed to determine whether multiple *wh*- fronting truly exists in the grammar as a productive multiple question formation strategy. The judgments reported in (26) were observed stably from late 2022 to summer 2023, leading to the initial conclusion that multiple *wh*- fronting exists in the language. However, in fall 2023, structures like those in (26b) were consistently judged ungrammatical, leading to our present reluctance to fully commit to the existence of multiple *wh*- fronting in the grammar. The third author of this paper (our language consultant) speculates that the availability of multiple *wh*- fronting may have an inter-generational dimension, reporting that older Ekhwa Adara speakers like his mother or grandparents seemingly allow/produce multiple *wh*- fronting structures. If all three multiple *wh*- question formation strategies are indeed available in the grammar, strategies (i) and (ii) are clearly preferred over

strategy (iii). Of relevance to the question of whether multiple *wh*-fronting truly exists, we note that multiple focus movement does not seem to be available in declarative sentences, as shown in (27).

- (27) a. * ɔɲfó_i ku ijâ_j ɲu omúsé sú mani —_j —_i
 song FOC Ija FOC Omuse FUT teach
 Intended: ‘It is a SONG that Omuse will teach IJA.’
 b. * ijâ_i ɲu/ku ɔɲfó_j ku omúsé sú mani —_i —_j
 Ija FOC song FOC Omuse FUT teach
 Intended: ‘It is a SONG that Omuse will teach IJA.’

In order to emphasize more than one non-interrogative argument in the clause, the preferred strategy is to leave one item in-situ with a change in intonation (represented by all caps), as shown in (28).²¹

- (28) ɔɲfó ku omúsé sú mani IJÂ
 song FOC Omuse FUT teach Ija
 ‘It is a SONG that Omuse will teach IJA.’

The absence of multiple focus-fronting in this context potentially casts doubt on the existence of multiple *wh*-fronting in the language (or at least in the grammar of the third author/others of his generation). If it does exist, something we leave for future research, then Ekhwa Adara is exceptional in its behavior regarding multiple *wh*- questions, as it employs a multiple fronting strategy that is not found, as far as we know, in any other African language.

4 Conclusion

We summarize our key findings on question formation in Ekhwa Adara as follows. Polar questions in Ekhwa Adara are primarily formed using final-vowel lengthening and L% boundary tone insertion. Furthermore, they are optionally marked by the sentence-initial Q particle *kó*. The only exception to this rule is found in embedded polar questions, which require the presence of an embedded left peripheral Q particle and the absence of final-vowel lengthening + L%

²¹A reviewer noted that the change in intonation in (28) may serve a different focus function besides the emphasis of a non-interrogative argument as documented here (for example, marking an exhaustive argument). We acknowledge this point and leave the exploration of this possibility for future research.

boundary tones. Regarding movement, Ekhwa Adara is an optional *wh*- movement language. When there is movement, the *wh*- item obligatorily precedes a focus marker, which can also optionally be used with in-situ *wh*- phrases. There is a selectional dependency between the *wh*- item and the accompanying focus marker in all contexts. Ekhwa Adara allows for long-distance *wh*- movement as well as simple (naked) partial *wh*- movement. Indirect questions in the language are formed via relativization, except when embedded under the verb ‘ask’, which involves *wh*- movement to the left periphery of the embedded clause. Ekhwa Adara potentially has three distinct strategies for forming multiple *wh*- questions: (i) all *wh*- items remain in-situ; (ii) one *wh*- item is fronted to clause-initial position, while the other(s) remain(s) in-situ; and (iii) multiple *wh*- fronting, a phenomenon that to the best of our knowledge is (currently) unattested in any other African language. Multiple *wh*- movement, to the extent that it truly exists in the grammar, is constrained – the initial fronted *wh*- item must be hierarchically inferior to the other(s), an effect we have characterized as an anti-superiority effect. Despite the existence of the anti-superiority effect in multiple *wh*- fronting constructions, *wh*- movement in Ekhwa Adara is not constrained by the superiority condition that limits *wh*- movement in languages like English.

Our investigation into Ekhwa Adara question syntax is ongoing. Future work will address the following issues (among others) related to the content of this paper: a fuller description of the distribution of optional Q particle *kó*; a better understanding of the various focus particles and the selectional dependency between them and *wh*- items; and an assessment of whether multiple *wh*- fronting truly exists in the grammar and if so, a structural analysis of multiple *wh*- fronting that derives the fact that the initial fronted *wh*- item must be hierarchically inferior to the other(s), given the absence of superiority effects in the language. We hope our description of the Ekhwa Adara interrogative system inspires further research on the grammar of this considerably under-documented language.

Abbreviations

1	first person	NOM	nominative
3	third person	PFV	perfective
ACC	accusative	PROG	progressive
COMP	complementizer	Q	question morpheme
DEF	definite	REL	relative marker
FOC	focus	SG	singular
FUT	future	SP	speaker

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References

- Aboh, Enoch. 2004. *The Morphosyntax of complement-head sequences: Clause structure and word order patterns in Kwa*. New York: Oxford University Press.
- Adesola, Oluseye. 2005. *Pronouns and Null Operators: A-bar Dependencies and Relations in Yoruba*. Rutgers University. (Doctoral dissertation).
- Adesola, Oluseye. 2006. On the Absence of Superiority and Weak Crossover Effects in Yoruba. *Linguistic Inquiry* 37. 309–318.
- Amaechi, Mary & Doreen Georgi. 2019. Quirks of Subject (Non-)extraction in Igbo. *Glossa: a Journal of General Linguistics* 4(1), 69. 1–36. DOI: <https://doi.org/10.5334/gjgl.607>.
- Bawa, Emmanuel. 2023. *The Distribution of Hausa Ko and its Influence in Adara*. Talk presented at Annual Conference on African Linguistics 54. University of Connecticut. June 14.
- Cahill, Michael. 2012. Polar Question Intonation in Kɔnni. In Michael R. Marlo, Nikki B. Adams, Christopher R. Green, Michelle Morrison & Tristan M. Purvis (eds.), *Selected Proceedings of the 42nd Annual Conference on African Linguistics*, 90–98. Somerville: Cascadilla Proceedings Project.
- Cahill, Michael. 2015. Polar Question Intonation in Five Ghanaian Languages. In Ruth Kramer, Elizabeth C. Zsiga & One Tlale Boyer (eds.), *Selected Proceedings of the 44th Annual Conference on African Linguistics*, 28–36. Somerville: Cascadilla Proceedings Project.
- Chomsky, Noam. 1973. *Conditions on Transformations*. Indiana Linguistics Club, University of Indiana, Bloomington.
- Chomsky, Noam. 1977. On Wh- movement. In Peter Culicover, Thomas Wasow & Adrian Akmajian (eds.), *Formal Syntax*, 71–132. New York: Academic Press.

- Devlin, Kerri, Blake Lehman, Travis Major & Harold Torrence. 2021. A Note on *Wh*-Questions in Avatime. In Akinbiyi Akinlabi, Lee Bickmore, Michael Cahill, Michael Diercks, Laura J. Downing, James Essegbey, Katie Franich, Laura McPherson & Sharon Rose (eds.), *Celebrating 50 Years of ACAL: Selected Papers from the 50th Annual Conference on African Linguistics*, 55–72. Berlin: Language Science Press.
- Fanselow, Gisbert. 2006. Partial Movement. In Martin Everaert, Henk van Riemsdijk, Rob Goedemans & Bart Hollebrandse (eds.), *The Blackwell Companion to Syntax*, vol. 3, 437–492. London: Blackwell Publishing.
- Hon, Luther, Grace Ajaegbu, Carol Magnusson, Uche S. Nweke & Zachariah Yoder. 2018. *A Sociolinguistic Survey of the Adara of Kaduna and Niger States, Nigeria*. (SIL Electronic Survey Reports: SIL International.
- Kandybowicz, Jason. 2004. Predicate Clefts, Derivations, and Universal Grammar. In Akin Akinlabi & Oluseye Adesola (eds.), *Proceedings of the 4th World Congress of African Linguistics, New Brunswick 2003*, 211–223. Köln: Rüdiger Köppe Verlag.
- Kandybowicz, Jason. 2008. *The Grammar of Repetition: Nupe Grammar at the Syntax-Phonology Interface*. Amsterdam: John Benjamins Publishing Company.
- Kandybowicz, Jason. 2017. On Prosodic Variation and the Distribution of *Wh*-In-situ. *Linguistic Variation* 17. 111–148.
- Kandybowicz, Jason. 2020. *Anti-contiguity: A Theory of Wh-Prosody*. New York: Oxford University Press.
- Kandybowicz, Jason, Bertille Baron, Philip T. Duncan & Hironori Katsuda. 2023. *Ikpana Interrogatives*. Oxford: Oxford University Press.
- Kandybowicz, Jason, Bertille Baron Obi, Philip T. Duncan & Hironori Katsuda. 2021. Documenting the Ikpana Interrogative System. *Journal of African Languages and Linguistics* 42. 63–100.
- Kandybowicz, Jason & Harold Torrence. 2015. The Prosodic Licensing of *Wh*-In-situ: Evidence from Krachi and Wasa. In Elizabeth Zsiga Ruth Kramer & One Boyer (eds.), *Selected Proceedings of the 44th Annual Conference on African Linguistics*, 146–157. Somerville: Cascadilla Press.
- Kuno, Susumu & Jane J. Robinson. 1972. Multiple *Wh* questions. *Linguistic Inquiry* 3. 463–487.
- Rialland, Annie. 2007a. Question Prosody: An African Perspective. In Carlos Gussenhoven & Tomas Riad (eds.), *Tones and Tunes: Studies in Word and Sentence Prosody*, 35–62. Berlin: Mouton de Gruyter.

- Rialland, Annie. 2007b. Question prosody: An African perspective. In Tomas Riad & Carlos Gussenhoven (eds.), *Typological studies in word and sentence prosody*, vol. 1, 35–62. Berlin: De Gruyter Mouton. <https://doi.org/10.1515/9783110207569.35>.
- Rialland, Annie. 2009. The African Lax Question Prosody: Its Realisation and Geographical Distribution. *Lingua* 199. 928–949.
- Rizzi, Luigi. 1997. The Fine Structure of the Left Periphery. In Liliane Haegeman (ed.), *Elements of Grammar*, 281–338. Dordrecht: Kluwer.
- Saah, Kofi. 1994. *Studies in Akan Syntax, Acquisition, and Sentence Processing*. PhD thesis, University of Ottawa.
- Schurr, Hagay, Jason Kandybowicz, Abdoulaye Laziz Nchare, Tysean Bucknor, Xiaomeng Ma, Magdalena Markowska & Armando Tapia. 2024. Absence of Clausal Islands in Shupamem. *Languages* 9(1), 7. 1–35. DOI: <https://doi.org/10.3390/languages9010007>.
- Simons, Gary F. & Charles D. Fennig. 2018. *Ethnologue: Languages of the World*. 21st edn. Dallas: Dallas: SIL International. <http://www.ethnologue.com>.
- Torrence, Harold & Jason Kandybowicz. 2013. Comparative Tano Interrogative Syntax. In Olanike Ola Orié & Karen W. Sanders (eds.), *Selected Proceedings of the 43rd Annual Conference on African Linguistics*, 222–234. Somerville: Cascadilla Press.
- Torrence, Harold & Jason Kandybowicz. 2015. *Wh- Question Formation in Krachi*. *Journal of African Languages and Linguistics* 36. 253–285.